

Linear Regression

June 2025

Why?

Your First Dataset

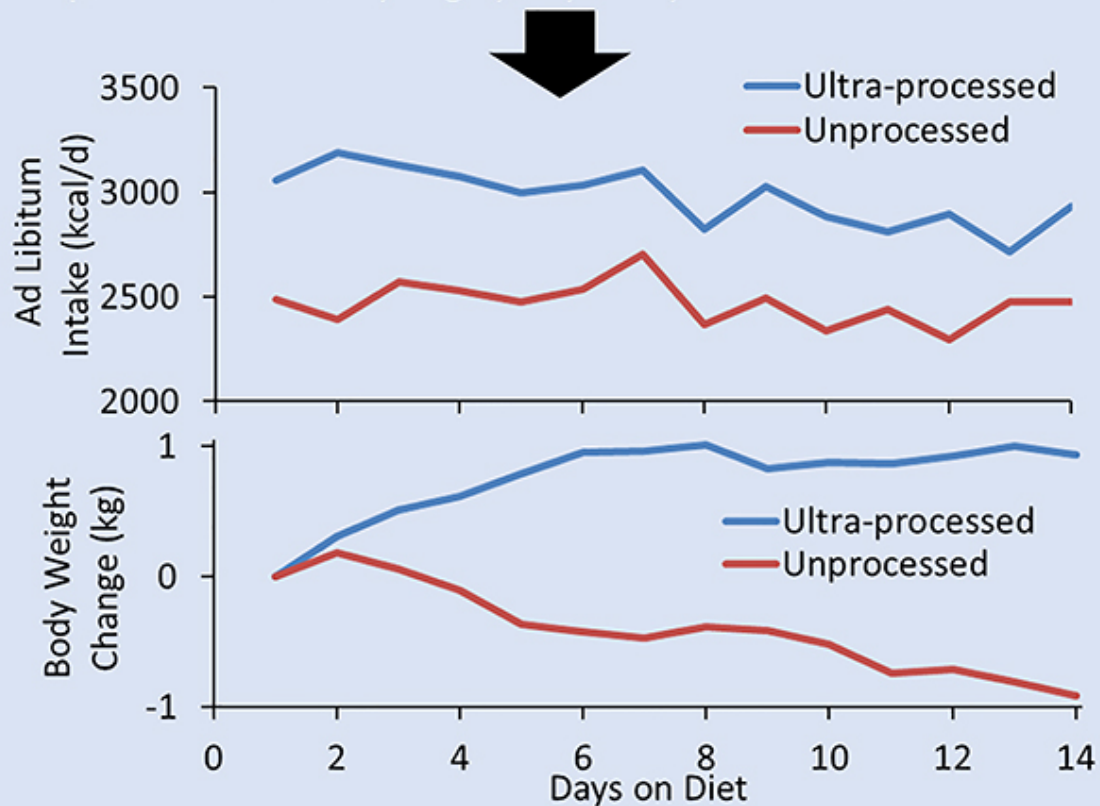
Ultra-processed Diet



Unprocessed Diet

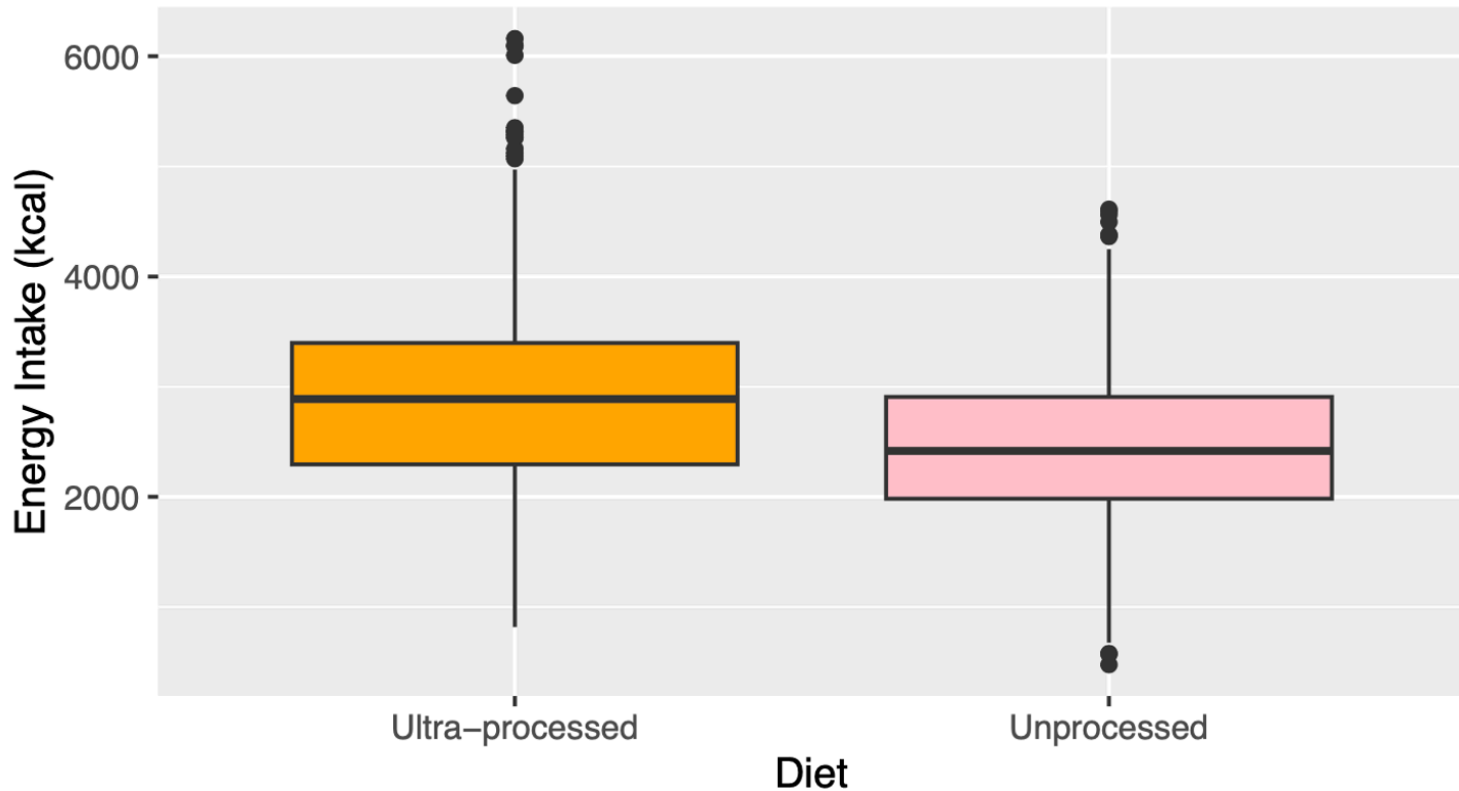


Diets were presented in random order and matched for provided calories, sugar, fat, fiber, and macronutrients

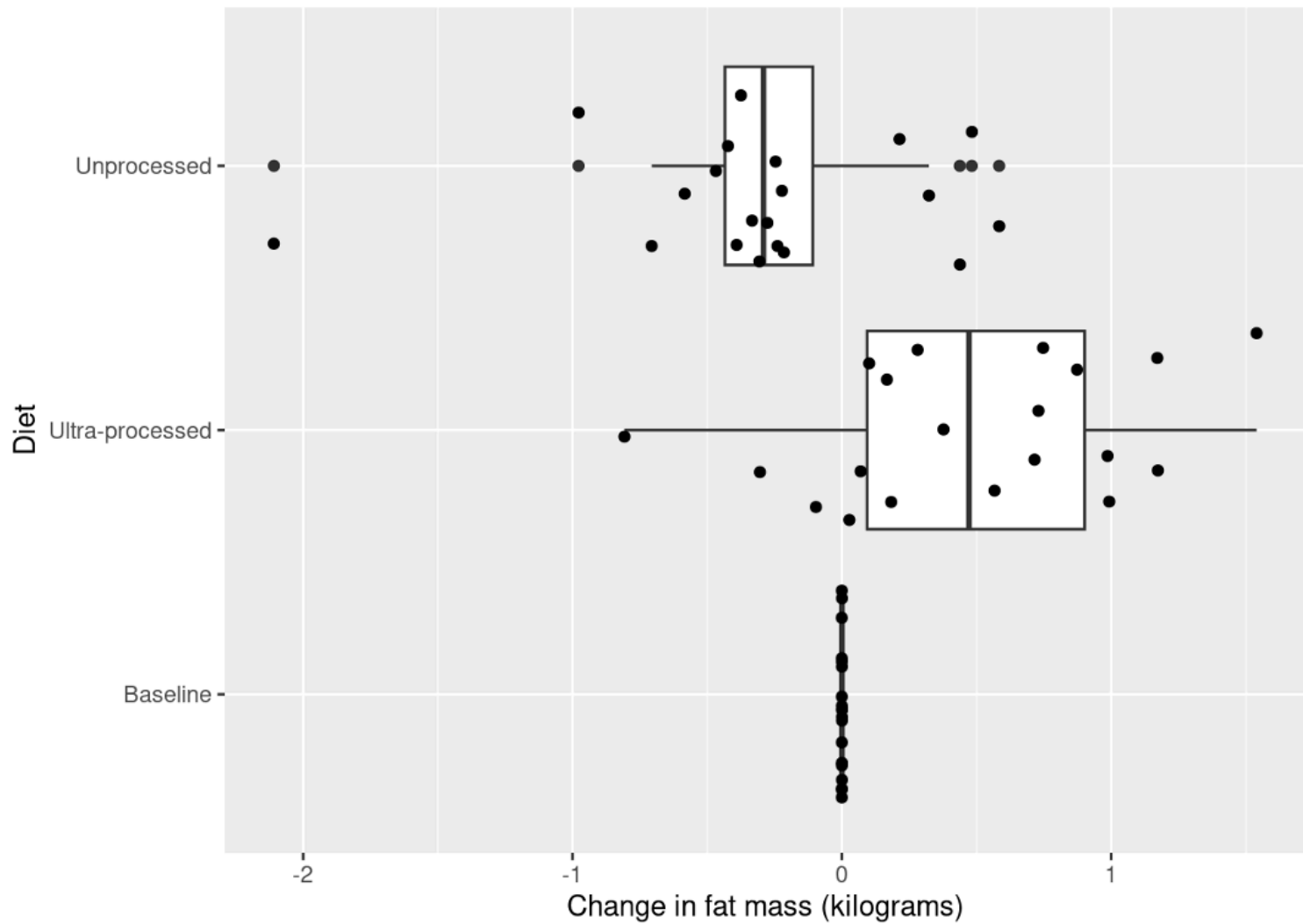


Or...the student versions

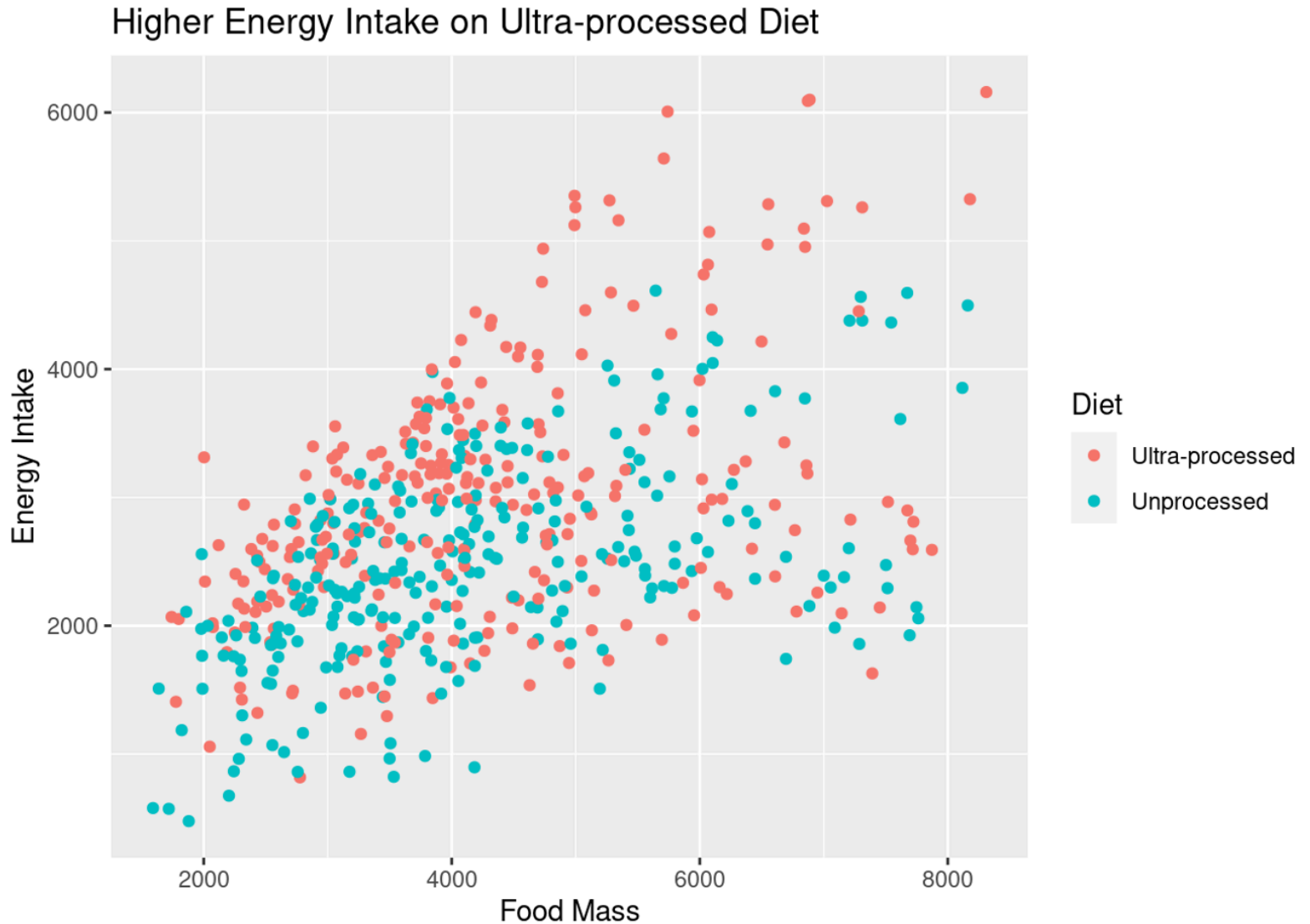
Ultraprocessed diets
exhibits higher energy intake than unprocessed diets



Or...the student versions



Or...the student versions



So, Why?

- Understand how *more than one thing* at a time influences our outcome
- *Quantify* which patterns are likely real and which might be just noise

Data

Facial width-to-height ratio is associated with agonistic and affiliative dominance in bonobos (*Pan paniscus*) (J.S. Martin and colleagues)

```
bonobos <-  
  read_csv(file='https://sldr.netlify.app/data/bonobo_faces.csv')
```


Facial Width-Height Ratio

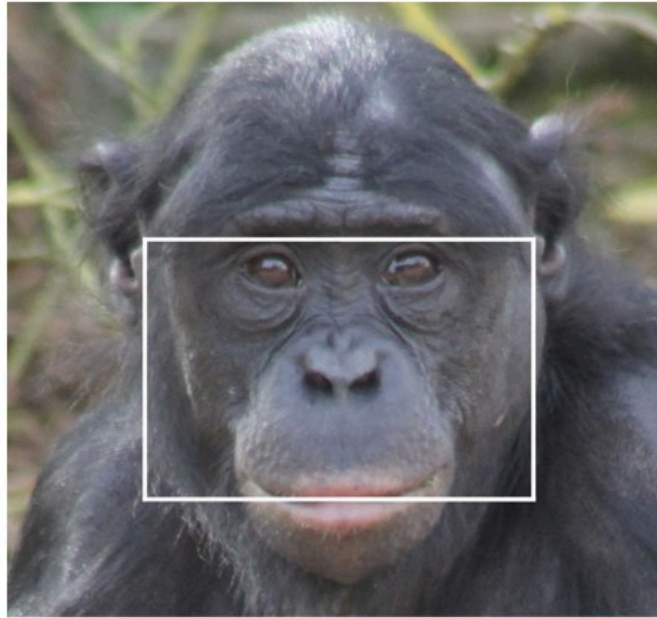


Figure 1. Facial width-to-height ratio. fWHR was measured by dividing the bizygomatic breadth (white box width) by the distance between the brow ridge and upper lip (white box height). The tragus was used for reference when facial hair covered the maximal cheek prominence.

Data: Variables

We will explore data - let's think a bit first.

- **Name** of the individual
- **Group**: zoo where the individual lives
- **Sex**, "Male or "Female"
- **Age** in years
- **fWHR**, facial width-height ratio
- **AssR**, assertiveness score
- **normDS**, another dominance score
- **weight** in kilograms

Simple Linear Regression

("Simple" means "one predictor")

$$y = ?$$

Choose Response

- Response is the variable of greatest interest
- Response is what you may want to *predict*
- Response may be causally dependent on predictor

Choose Predictor(s)

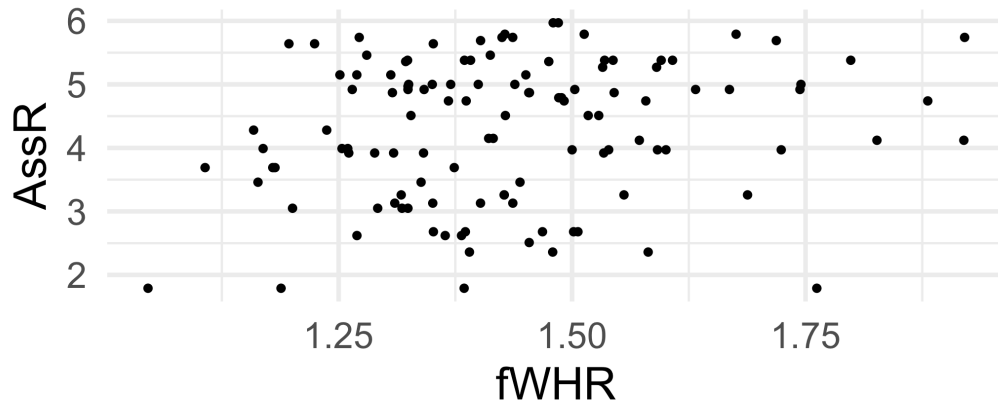
- Response may be harder to measure, and predictor easier
- (Should have data on predictor)
- **Choosing predictor that "looks best" in data -> biased inference**

For Bonobos:

- Response
- Predictor

Plan. STOP! Explore.

```
gf_point(AssR ~ fWHR, data = bonobos)
```



To fit model: `lm()`

```
m1_simple <- lm(AssR ~ fWHR,  
                 data = bonobos)
```


Print Results

Just Parameter Estimates

```
coef(m1_simple)
```

##	(Intercept)	fWHR
##	2.547350	1.213902

Print Results

Full Summary

```
summary(m1_simple)
```

Summary

```
##  
## Call:  
## lm(formula = AssR ~ fWHR, data = bonobos)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.8963 -0.8405  0.2307  0.8442  1.6485   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)   2.5474     0.8518   2.991  0.00341 **    
## fWHR           1.2139     0.5907   2.055  0.04213 *     
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 1.09 on 115 degrees of freedom  
## Multiple R-squared:  0.03542,    Adjusted R-squared:  0.02704   
## F-statistic: 4.223 on 1 and 115 DF,  p-value: 0.04213
```

(Part of the) Model Equation

$$y = _ + _ x \dots$$

But the Line isn't Perfect

There is some error ϵ in our model for the data, so we should keep track of it.

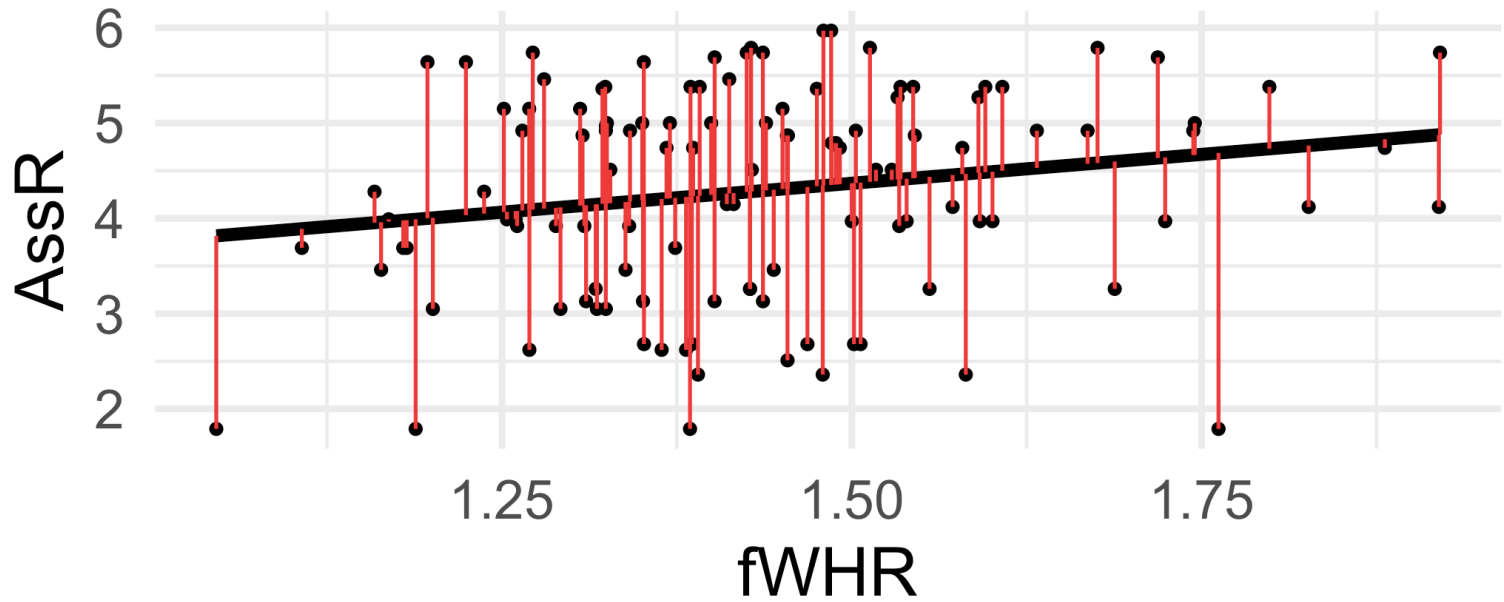
$$y = \beta_0 + \beta_1 x + \epsilon$$

ϵ is NOT *one number*

The amount of error is different for (almost) every data point!

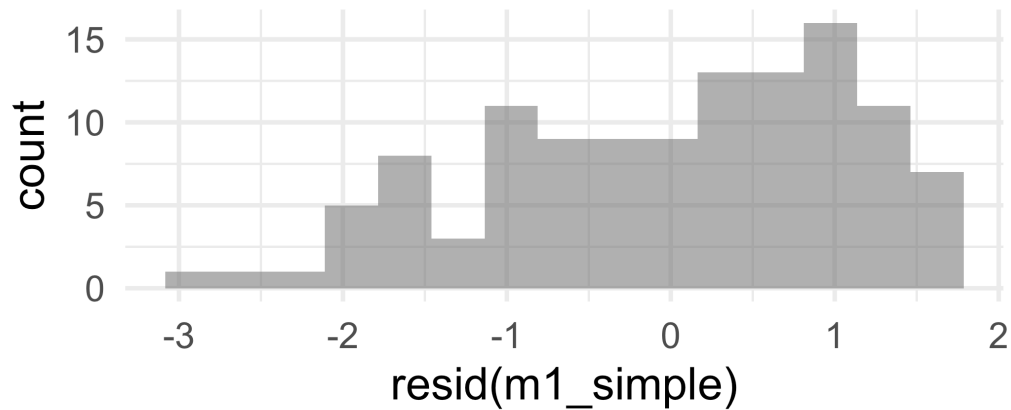
- How can we express this mathematically?

Regression Residuals = "errors"



Summarize: Distribution?

```
gf_histogram(~resid(m1_simple),  
             bins = 15)
```



Back to Summary

```
##  
## Call:  
## lm(formula = AssR ~ fWHR, data = bonobos)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.8963 -0.8405  0.2307  0.8442  1.6485   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)   2.5474     0.8518   2.991  0.00341 **    
## fWHR           1.2139     0.5907   2.055  0.04213 *     
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
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## Multiple R-squared:  0.03542,    Adjusted R-squared:  0.02704   
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```

Complete Regression Equation

$$y = \beta_0 + \beta_1 x + \epsilon,$$

where

$$\epsilon \sim N(0, \sigma)$$

Bonobo Regression Equation

$$y = \text{_____} + \text{_____}x + \epsilon,$$

$$\text{where } \epsilon \sim N(0, \text{_____})$$

Multiple regression

- Rarely does our response variable **really** depend on only one predictor.
- Can we expand our formulation to include more predictors? (Example: **normDS** also predicts **AssR**?)
- In R, it's super easy:

```
m2_2q <- lm(AssR ~ fWHR + normDS,  
             data = bonobos)
```

Summary + Equation

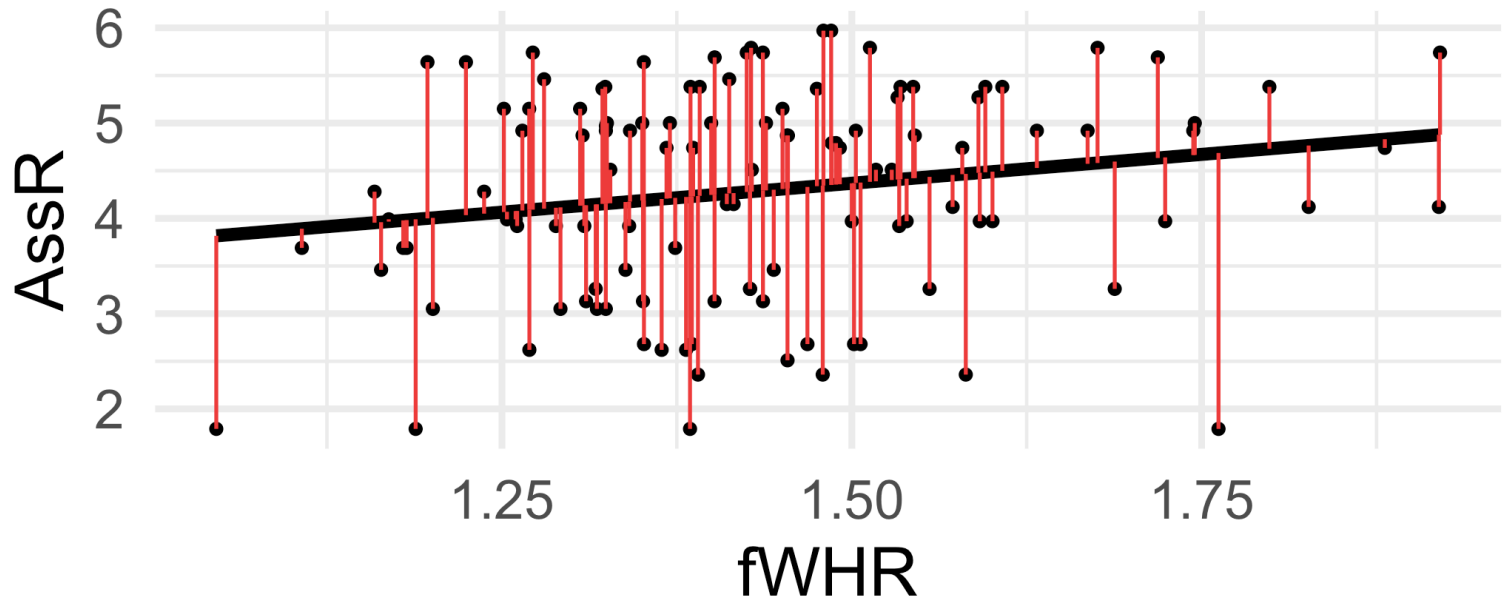
```
##
## Call:
## lm(formula = AssR ~ fWHR + normDS, data = bonobos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.9993 -0.7592  0.1832  0.8279  1.7172
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.53889     0.85610   2.966  0.00369 **
## fWHR          1.40331     0.62298   2.253  0.02622 *
## normDS       -0.09918     0.09687  -1.024  0.30810
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.094 on 113 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.04403,    Adjusted R-squared:  0.02711
## F-statistic: 2.602 on 2 and 113 DF,  p-value: 0.07855
```

Choosing Predictors (Again)

- Here: build simple -> complex *to show math machinery*
- In practice: **Think before you model**
 - Rule of thumb (from Harrell): $p < \frac{n}{15}$
 - p is number of parameters want to estimate; n is sample size (rows in data)

How Fitting Happened

Simple Linear Regression Residuals



Least Squares Estimation

Minimize:

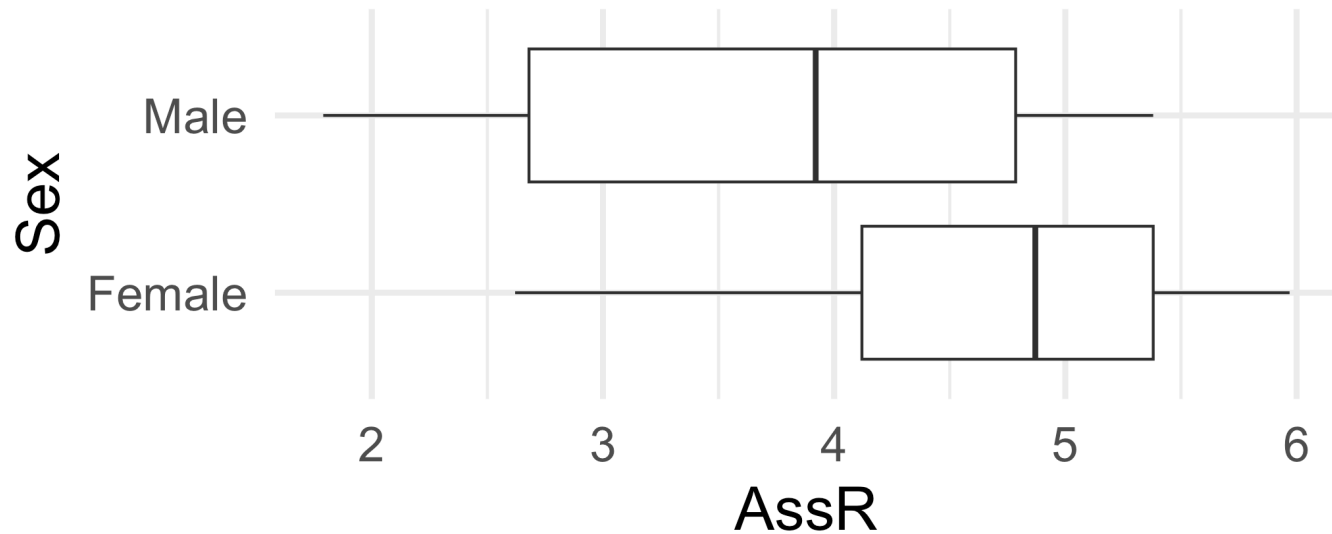
$$SSE = \sum_{i=1}^n e_i = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Multiple Predictors?

- Harder to draw
- Just as easy to compute $\hat{y}$...
- and thus compute the observed residuals e_i
- and the sum of squared residuals

See: <https://setosa.io/ev/ordinary-least-squares-regression/>

Predictors with 2 categories



Predictors with 2 categories

```
m3_2q1b <- lm(AssR ~ fWHR + normDS +  
              Sex,  
              data = bonobos)  
coef(m3_2q1b)
```

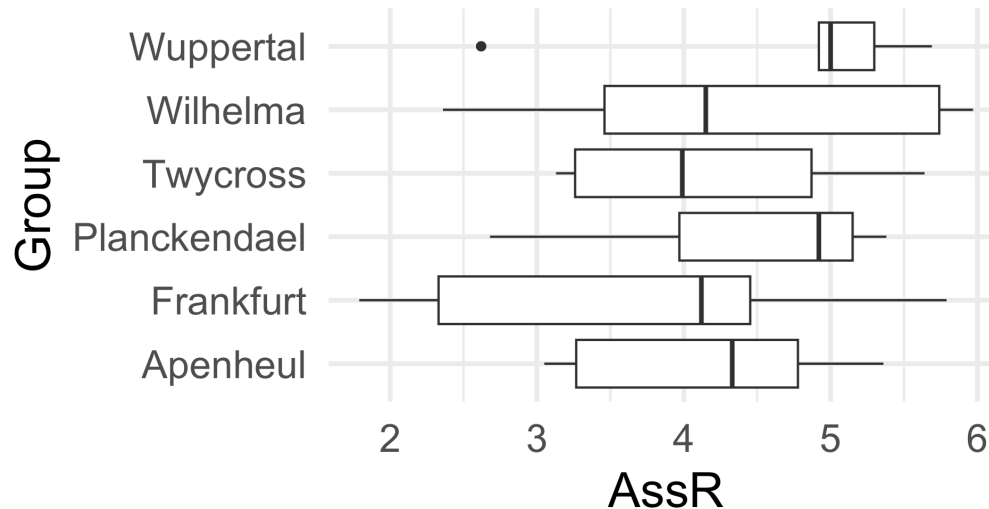
```
## (Intercept)          fWHR          normDS          SexMale  
##  2.07913144  1.89581129 -0.01849396 -1.11030054
```

Predictors with 2 categories - Summary, Equation

```
##  
## Call:  
## lm(formula = AssR ~ fWHR + normDS + Sex, data = bonobos)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.47788 -0.61852  0.09069  0.73386  1.59519   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  2.07913     0.75375   2.758 0.006786 **     
## fWHR         1.89581     0.55186   3.435 0.000831 ***    
## normDS      -0.01849     0.08592  -0.215 0.829962        
## SexMale     -1.11030     0.18681  -5.943 3.22e-08 ***    
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.9581 on 112 degrees of freedom  
## (1 observation deleted due to missingness)  
## Multiple R-squared:  0.2732,    Adjusted R-squared:  0.2538   
## F-statistic: 14.04 on 3 and 112 DF,  p-value: 7.86e-08
```

More categories

```
gf_boxplot(Group ~ AssR,  
            data = bonobos)
```



More Categories

```
m3_2q2c <- lm(AssR ~ fWHR + normDS +  
               Sex + Group,  
               data = bonobos)
```

More Categories: Summary, Equation

```
##
## Call:
## lm(formula = AssR ~ fWHR + normDS + Sex + Group, data = bonobos)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.5148 -0.5901 -0.0118  0.6610  1.5405
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.16779    0.79783   2.717  0.00768 **
## fWHR            1.65461    0.56485   2.929  0.00415 **
## normDS          0.07067    0.08782   0.805  0.42277
## SexMale        -1.23398    0.18576  -6.643 1.32e-09 ***
## GroupFrankfurt -0.61604    0.34951  -1.763  0.08083 .
## GroupPlanckendael 0.35141    0.31958   1.100  0.27398
## GroupTwycross    0.09313    0.30547   0.305  0.76105
## GroupWilhelma   -0.08112    0.33549  -0.242  0.80940
## GroupWuppertal   0.47304    0.32545   1.453  0.14901
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9222 on 187 degrees of freedom
```

Predictions by Hand

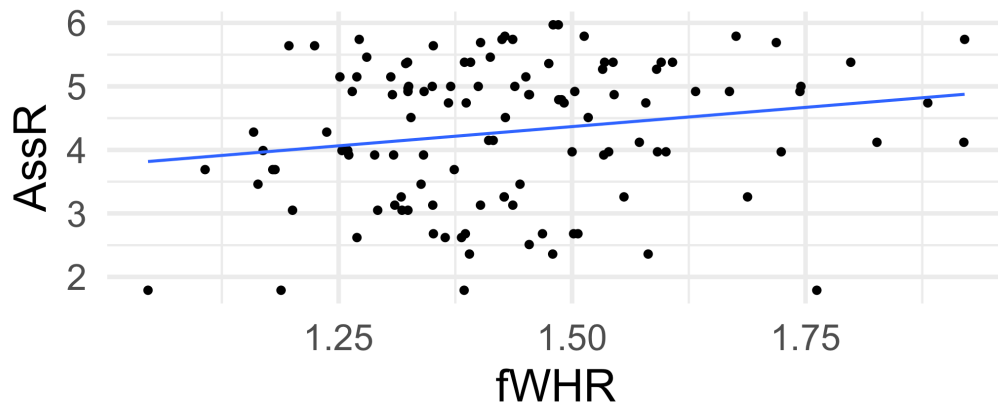
What is the expected $ASSR$ (according to this model) for 30 kg female bonobos at the Wilhelma zoo with $fWHR$ of 1.5 and $normDS$ of 2.5?

`gf_lm()` **or** `geom_lm()`:

NEVER Do This...

well, almost never -- why?

```
gf_point(AssR ~ fWHR, data = b2) |>  
  gf_lm()
```



Feedback

<https://cutt.ly/arm-feedback>

